

1

The tree, and the one rule

Five leaves, and each one is a lever with its own bill. Branches multiply, so two 10 percent lifts give 21 percent, and 15 percent off for 10 percent more volume leaves 0.935 of today.

Crux. Write every branch as a numerator over a denominator before any number is computed, and move a budget only after finding which branch is short.

2

Every leaf is a metric with a bill

Leaf	Over what	What moving it costs
Customers	A count, in the window	Marketing; buyers who never return
Orders per customer	Distinct customers, same window	Loyalty; paying those who would return
Items per order	Orders	Merchandising; low-margin baskets
Price per item	Items	Volume; the price-sensitive leave
Discounts	Revenue before discounts	Margin, traded for quantity

3

One word, four readings

Booked, not cancelled, delivered, and after discounts: four honest totals of the same orders, each answering a different question.

Crux. Name the definition and the window before the number: "Revenue, all booked orders, 1 July to 26 September".

4

The accumulator, three ways

```
revenue = 0                                # start, before the loop
for order in ORDERS:
    revenue += int(order["amount"])         # update, once per record
print(revenue)                             # finish, after the loop
```

Count adds 1. Sum adds the value. A distinct count appends an id only when it is `not in` the list. A filter puts an `if` before the update.

5

Mean or median

The number is for	Report
A typical order	The median
A total that must add up	The mean
A first look at a file	Both, and the gap

The median of an even count is the average of the two middles: indexes `n // 2 - 1` and `n // 2`. One value moved by `d` moves the mean by `d over n` and the median not at all.

Crux. When the mean sits far above the median, the gap is the finding: read the top of the sorted list and name what sits there.

6

Errors met today, read from the last line up

Last line	First move
<code>NameError: name 'ORDERS' is not defined</code>	Restart and Run All
<code>TypeError: ... 'int' and 'str'</code>	Print the record the loop stopped on
<code>KeyError: 'Amount'</code>	Copy the key from the record
<code>ValueError: invalid literal for int()</code>	Decide the rule for that field

7

The sentence to Meera

Claim, the branch to examine first. Evidence, a number with its definition and window. Caveat, what one window cannot show. Next step, the comparison that settles it.

Crux. A sentence without its evidence is an opinion, and one without its caveat is a promise.

THE WORDS ON THIS SHEET

- Revenue tree** Revenue drawn as the counts and prices that multiply into it, less discounts
- Definition** The words that say which orders a total counts and over which dates
- Kernel** The Python process that remembers the cells it ran since its restart
- Mean** The total spread evenly over the count
- Denominator** What a rate is divided by, said aloud before it is computed
- Accumulator** A total started before a loop, updated once per record, read after it
- Median** The middle of the sorted values, or the average of the two middles
- Skew** A few values far above the rest, pulling the mean from the median